MuRET: A music recognition, encoding, and transcription tool

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ABSTRACT

The transcription process from early and modern notation manuscripts to a structured digital encoding has been traditionally performed following a fully manual workflow. At most it has received some technological support in particular stages, like optical music recognition (OMR) of the source images, or transcription to modern notation with music edition applications. Currently, there is no mature and stable enough solution for the OMR problem, and the most used music editors do not support early notations, such as the mensural one. In this work, a new tool called *MUsic Recognition, Encoding, and Transcription* (MuRET) is introduced, which covers all transcription phases, from the manuscript source to the encoded digital content. MuRET is designed as a technology-focused research tool, allowing different processing approaches to be used, and producing both the expected transcribed contents in standard encodings and data for the study of the transcription process itself.

KEYWORDS

Notation transcription, Encoding, Optical music recognition

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1 INTRODUCTION

Optical music recognition (OMR) systems are key tools for publication of music score collections that are currently found only on paper. The main goal of an OMR system is to convert music sheet images to a digital structured format, like Music Encoding Initiative (MEI) or MusicXML. Under the perspective of the cultural heritage preservation, many of these collections are handwritten material, and some of them can be found in early notation systems. State-of-the-art OMR systems are not ready yet to operate in an automatic, unsupervised way in these contexts, so the intervention of a human expert is a must in the recognition and edition process [9]. Therefore, it is necessary to consider interactive paradigms, and we have to pay attention on the human-machine interaction aspects related to this task and how to optimize the expert user feedback.

In particular, Spanish white mensural notation was the dominant code for writing music in Spain during the 16th to 18th centuries, producing large collections of documents yet to be made accessible to the public. In all these cases, just the scanned or photographed images of the scores are available, and not their transcription into symbolic music formats. The OMR techniques available today are far to be ready for being applied to these kind of handwritten documents. Thus, one of the main goals of the *MUsic Recognition, Encoding, and Transcription* (MuRET) project is to be able to process this kind of documents, covering the workflow from the digitized images to the production of the digital symbolic format.

The proposed system may be able to produce both a transcribed copy of the original (*diplomatic* version) and the material for elaborating a *critical* edition, where the possible errors or mistakes in the original work are corrected and solved. This is not a trivial task. In addition to possible problems that the original scribe might have made, the papers usually contain stains, blots or even holes that may affect the original material. Also, the *semitonia subintelecta* [1] issue leads to missing musical elements in some manuscripts that might be indicated in the critical edition for improving the readability. One important feature of MuRET is the possibility to render also a translated version to a modern notation score, readable by a contemporary musician, and allowing the public to enjoy and search into the digital contents of these works, either as a musicologist or as a musician.

From the technical point of view, it is important to note that the symbol recognition and processing stages of the system are designed using machine learning (ML) techniques. Most available systems are based on heuristic rule methods, but, as stated in [3], one of the problems of OMR is the existence of too many repertoire-dependent context rules. Therefore, it is hard for them to be extended to recognize early music notations, that are different from the modern one. By applying innovative ML technologies we can build models based on *training pairs*: sets of data examples that are presented together with a ground-truth label. This is a key point of this method, because it opens up the possibility of adapting the system, in principle, to any music notation, provided labelled data is available for the system to learn. In fact, the proposed system is being developed working on two different collections of handwritten music scores: vocal music manuscripts, written in Spanish mensural notation, and Spanish traditional songs transcribed by hand in modern music notation by ethno-musicologists.

[1] Accidentals are not always precisely indicated in the manuscript

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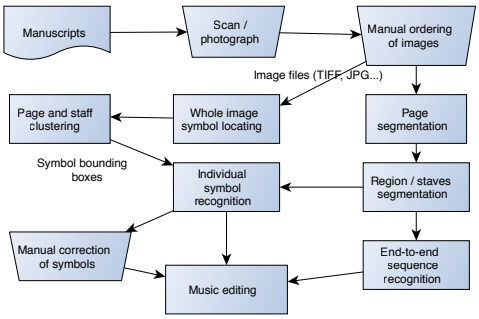


Figure 1: Allowed workflows in MuRET

2 METHODOLOGY

Typical OMR systems such as Audiveris[2] or Photoscore[3] are mainly devised to extract the musical content from printed or manuscript score sheets in order to edit them further in music edition applications. Aruspix[15] goes a step beyond and allows the superimposition and the collation of early music prints. However, none of them is devised with the requirements of being able to recognize the contents of printed or manuscript scores from different approaches, allowing the manual introduction of new materials, the encoding into all current standards, and the assisted transcription into a edited version ready for preparing a critical edition.

MuRET has been designed to fulfil these requirements, giving support to the whole process of transcription of a manuscript source in such a way the traceability of each step is maintained while allowing a number of OMR approaches (detailed below). In addition, as this system is conceived as a research tool, it has been equipped with features to measure the efficiency and effectiveness of the different user and machine tasks and approaches.

The scheme in Fig. 1 describes the different automatic and manual operations performed in the system for any of the considered approaches. The different workflows are implemented in such a way that the process can be accomplished in an arbitrary number of working sessions and users. Also, the stages include the logging of user actions for future performance assessment.

In the first stage the system is fed with digitised image files of the manuscripts. Then, the user has to organize the images according to the books, and their order within them. This manual organization could be replaced in the future by an automatic or assisted one. Afterwards, the OMR process can be carried out under different approaches. Further details about OMR approaches are given in Section 3.

In all cases, our OMR operations generate two kind of sequence symbol encodings [5]. One is named *agnostic*, which contains symbols by their graphic value and their position, without analysing the role of the symbol in the score, *e.g.*, a ♯ symbol is tagged as a sharp accidental, not as note modifier or as an element of a key signature. This way, a D major key signature is encoded as two sharps in the positions where they are found. The other encoding is the *semantic* one, where meaningful musical information is encoded in any standard music format. Here, the D major key signature is encoded as an explicit D Major key code. The conversion from the agnostic encoding to a semantic one can be accomplished in a relative straight-forward way [2] in the case of simple monodic music in common modern western notation (CWMN). The case of mensural music requires a more specific procedure to deal with context-dependent interpretations of graphical symbols [17].

Furthermore, MuRET is designed under the assumption that no OMR approach is able to achieve a perfect performance. Thus, the user intervention is required. This assessment and correction task can be accomplished at graphical symbol level (agnostic) or by editing the music content obtained directly (semantic). MuRET allows both of them. In the case of choosing the correction of the agnostic encoding, symbols can be amended using short-cuts and tool-bars, where the user can select the expected symbol and position in the staff.

This possibility of correcting the output from a graphical point of view allows us to distribute tasks depending on their complexity. In non-standard notations, such as manuscript mensural one, an expert musicologist is required for a correct interpretation of some musical symbol information, like, for example, their perfection or imperfection [1]. Many times, the musical output obtained from the system is incorrect because it has misclassified a graphical symbol, *e.g.*, a sharp by a brevis. With this kind of system mistakes, the expert user time is wasted in correcting things that a non-user expert could do. This is the aim of the optional agnostic graphical correction phase.

Finally, once the symbols are corrected, either from an agnostic or semantic approach, music content is further edited introducing aspects such as editorial accidentals due to the *semitonia subintelecta* missing alterations, or missing musical elements in some handwritten manuscripts such as the initial clefs of a new system.

3 OPTICAL MUSIC RECOGNITION

From our experience, we believe that it is difficult to find a single approach for OMR that can be effective in all kinds of situations. Therefore, we implement in MuRET three types of automatic recognition, each of which analyses the image at different levels of globality:

User-driven symbol recognition: the first approach is based on the work of Calvo-Zaragoza et al. [6], where the user locates the symbols manually. Typically, the computer mouse can be used to indicate the position of the symbols, but we also allow in MuRET the use of a digital pen, resulting in a more ergonomic interaction (see Fig. 2). Either way, this procedure yields a bounding box for each symbol, that is received by a classifier as input. The output is the category prediction of the symbol therein. Another classifier is then used to estimate the vertical position of the symbol within the staff. By clustering the positions of the bounding boxes on the y-axis, the number of staves can be determined, while an ordering on the x-axis allows determining the actual order in the sequence of symbols. The advantage of this approach is that the classifiers need little ground-truth data to obtain good results, but users' effort is demanding. An example of this approach is depicted in Fig. 3.

Holistic staff recognition: in this approach, each staff is processed holistically, that is, without previous symbol segmentation (see

[2] https://github.com/Audiveris
[3] www.neuratron.com/photoscore.htm

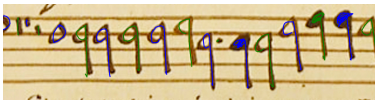


Figure 2: Manual tracing of symbols.



Figure 3: User-driven symbol recognition. Classifiers deal with manually located symbol bounding boxes.

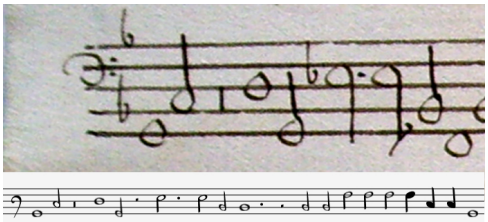


Figure 4: Staff-based end-to-end approach. Classifiers receive a entire staff region and yield a sequence of classified symbols, including their positions within the staff.

Fig. 4). This can be achieved by using continuous models like hidden Markov models [7] or recurrent neural networks [4, 5]. In this holistic approach, the categories predicted include both the shape of the symbol and its position on the staff. The advantage of these approaches is that the ground-truth data consists simply of pairs of staff images and their corresponding sequence of music symbols, without the need to indicate any geometrical information, and thus being less expensive to obtain. The separation of the pages into staves can be done heuristically, given that the staff lines provide an excellent reference for such purpose, or using ML techniques [8].

Full page recognition: the last approach we consider is a neural model that is applied directly to a full page [12]. In this case, given the image, the model returns a list of predictions, each of which indicates a bounding box and the category of the object therein (see Fig. 5). Note that this approach is equivalent to the first one, with the difference that the user does not have to manually locate the symbols. This approach has reported promising results for OMR [13, 14], although acquiring its ground-truth data is quite demanding.

Note that these approaches work at the graphical level (agnostic representation), ignoring some semantics of the music notation itself that can be recovered in a post-processing stage. For example, in the case of notes, the vertical position within the staff, along with the clef and eventual alterations, allows inferring their pitches.

Furthermore, it is likely that when starting a new manuscript, the user-driven approach may be the best option because, in spite

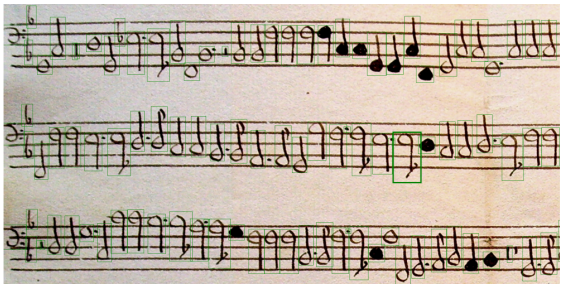


Figure 5: Identified symbols in the whole page recognition approach.



Figure 6: Spanish late white mensural transcription and translation to modern notation.

of requiring the manual localization of each symbol, the accuracy may be higher from the beginning. As ground-truth data is obtained from the manuscript itself, the other approaches could be appropriately trained. This is why we believe that MuRET will represent an excellent framework over which to evaluate OMR algorithms.

Our intention is also to take advantage of MuRET infrastructure to carry out user-centred studies, which will give insights concerning which of these approaches is the most effective in each situation. The ultimate goal is to obtain the structured encoding of the music sources with the least possible effort from the user. The effort of the user is closely related to the accuracy of the recognition models (the higher accuracy, the less corrections are necessary) but also with the type of corrections and interactions required. That is why carrying out this type of studies is important.

4 MUSIC EDITING

As mentioned above, one of the fundamental objectives of MuRET is to obtain a transcription of the manuscript without any ambiguities, so that both the musicologist and the publisher can generate a critical edition from it. This stage includes two tasks: the optional correction of the recognized graphical symbols from the OMR, and the correct assignment of musical functions to those graphical symbols, such as distinguishing a division or prolongation dot in mensural notation.

In the case of the transcription of common western notation manuscripts (note that currently MuRET is focused on the transcription of monodic melodies from a traditional Spanish repertoire), the user can edit directly the recognized content without any other help. However, in the case of mensural notation, it is very useful to be able to edit the mensural content by simultaneously assessing the correctness of the transcribed modern version (see Fig. 6).

4.1 Score rendering

MuRET does not intend to be a score editing application, but it must fulfil the needs to any editing and encoding need the musicologist may have. Thus, our aim has been set to easily allow all these needs for the considered corpora, that will be extended in the future.

The typography "Patriarca" has been developed on purpose, extending "Solera", introduced in [17], for allowing an easy editing of mensural manuscripts. Although there exists some mensural fonts that could have been used, such as Bravura [4], they do not contain all the glyphs that appear in the manuscripts we are dealing with. Besides, in the process of manually assessing the correction of the OMR output, if the digital font resembles the original calligraphy, the user effort to compare symbols is substantially reduced. Bravura contains typographic fonts and not calligraphic ones.

Furthermore, the current goal of MuRET is not to generate final edited preprints, but to produce contents to be sent to online services [11, 16, 19] or publishers. As above mentioned, we can give as granted that what MuRET exports in any interchange format will be edited by these publishers to fulfil their publishing workflow. Thus, a printed image must be generated with the aspect of how approximately the transcription should look like. MuRET exports this kind of output using PDF.

5 ENCODING

There are several factors that have prevented OMR technology from a faster progress during many years. One of them has been the long-lasting lack of shareable common encoding formats, more in the internals of the system than in the output, for which formats such as NIFF [18] were proposed. To the best of our knowledge, only Audiveris and Sharpeye have made public their internal encoding formats. As it could be expected, they do not share a common structure. Regarding the output generated by this kind of systems, most of them export MusicXML, other ones render NIFF, and only Aruspix produces MEI.

Even though MuRET can import and export using several standard formats that will be discussed below, we do not rely on them to encode all the elements considered in the whole recognition and transcription process because no one is able to do it. Thus, as almost any software application, we use an internal ad-hoc format to do it. In order to be as transparent as possible, we are using an XML based format whose schema will be published in the HISPAMUS project website [5] to allow other systems to read and use any intermediate result of the tool. We have based it on systems such as SharpEye (.mro format) [6], and Audiveris XML encoding, and expanded to allow capabilities not considered in those systems, such as the encoding the output of staff based end-to-end classifiers, or the agnostic and semantic encodings [4].

5.1 Interchange formats

When converting digital images into digital scores there are two factors that must be taken into account. Maybe, the most important, is the data sustainability [10] for its preservation along time. A very good system may be useless if the content it produces is stored in a non standard encoding. The second factor, related to the former, is how to transmit this content both to online services allowing utilities such as searching, and to publishers that must process the encoded music in order to created critical editions.

Currently, there is not a unique universal format accepted by all publishers and online service applications. From the commercial tools point of view, the most extended interchange format is MusicXML. This format, however, cannot encode contents such as mensural notation, the critical apparatus, some editorial markups, or many metadata contents required for being correctly catalogued by digital libraries. MEI fills these needs, however, being an evolving format driven by the research community, it is slightly being modified in the last years, and in some cases, either it allows different ways of encoding the same content, or it lacks from some features required by our system. MEI has a protocol to introduce new proposals suggested by any researcher, and the possibility of being customized. Some online music digital libraries currently use Lilypond as encoding format for its score rendering quality and wide capabilities. However, this cannot be used as an interchange format, but as a final way of editing and digitally publishing the scores. Projects such as the Josquin Research Project [7] use `**kern` as its base encoding format, mainly focused on analysis purposes.

In this context, our approach is to avoid our system to be based on any of these formats, but to use our own object oriented hierarchy [8] to represent any of the contents to be encoded, and export (and in some cases import) to (from) those formats. Currently MuRET is exporting to MEI, MusicXML, Lilypond, and `**kern`. In the exporting process, the contents that cannot be exported in the chosen format are shown in a human readable text file for avoiding losing information. When exporting to MEI, the customization mechanism is used to export features not supported yet by the language. Note that our aim is to adhere to standards, thus, any extension will be proposed in the work groups of MEI as a suggested improvement.

6 CONCLUSIONS

A new tool named MuRET (*Music Recognition, Encoding, and Transcription*) has been introduced, which is designed to be used in both written music transcription projects and OMR research. This tool is currently focused on two specific repertoires, monodic melody handwritten scores containing traditional Spanish music, and white mensural notation from the 16th to 18th centuries from handwritten manuscripts. Nonetheless, it has been conceived to be easily extended to any kind of repertoire. The tool itself, the programming libraries supporting it, and the OMR technology used, are currently under research and development, and they are expected to be totally usable by any end user in open source format in the upcoming years.

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[4] https://www.smufl.org
[5] At http://grfia.dlsi.ua.es/
[6] http://www.visiv.co.uk/tech-mro.htm
[7] http://josquin.stanford.edu
[8] https://github.com/davidrizo/im3

REFERENCES

[1] Willi Apel. 1953. The notation of polyphonic music, 900-1600. (1953), 1–532.

[2] Jorge Aracil Pérez. 2018. *Modelos computacionales para la transcripción musical.* Master's thesis. Universidad Internacional de La Rioja.

[3] Donald Byrd and Jakob Grue Simonsen. 2015. Towards a Standard Testbed for Optical Music Recognition: Definitions, Metrics, and Page Images. *Journal of New Music Research* 44, 3 (2015), 169–195.

[4] Jorge Calvo-Zaragoza and David Rizo. 2018. Camera-PrIMuS: Neural End-to-End Optical Music Recognition on Realistic Monophonic Scores. In *19th International Society for Music Information Retrieval Conference.* (in press).

[5] Jorge Calvo-Zaragoza and David Rizo. 2018. End-to-End Neural Optical Music Recognition of Monophonic Scores. *Applied Sciences* 8, 4 (2018), 606–629.

[6] Jorge Calvo-Zaragoza, David Rizo, and José Manuel Iñesta. 2016. Two (Note) Heads Are Better Than One: Pen-Based Multimodal Interaction with Music Scores. In *17th International Society for Music Information Retrieval Conference.* 509–514.

[7] Jorge Calvo-Zaragoza, Alejandro Héctor Toselli, and Enrique Vidal. 2017. Handwritten Music Recognition for Mensural Notation: Formulation, Data and Baseline Results. In *14th IAPR International Conference on Document Analysis and Recognition.* 1081–1086.

[8] Vicente Bosch Campos, Jorge Calvo-Zaragoza, Alejandro Héctor Toselli, and Enrique Vidal-Ruiz. 2016. Sheet Music Statistical Layout Analysis. In *15th International Conference on Frontiers in Handwriting Recognition.* 313–318.

[9] Liang Chen and Christopher Raphael. 2016. Human-Directed Optical Music Recognition. In *Document Recognition and Retrieval XXIII, San Francisco, California, USA, February 14-18, 2016.* 1–9.

[10] Reinier de Valk, Marnix van Berchum, and Peter van Kranenburg. 2017. Music encoding, formats, and data sustainability. Paper presented at the Music Encoding Congerence, Tours (France).

[11] Reinier de Valk, Anja Volk, Andre Holzapfel, Aggelos Pikrakis, Nadine Kroher, and Joren Six. 2017. MIRchiving: Challenges and Opportunities of Connecting MIR Research and Digital Music Archives. In *Proceedings of the 4th International Workshop on Digital Libraries for Musicology (DLfM '17).* ACM, New York, NY, USA, 25–28. https://doi.org/10.1145/3144749.3144755

[12] Ross Girshick, Jeff Donahue, Trevor Darrell, and Jitendra Malik. 2014. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition.* 580–587.

[13] Alexander Pacha and Jorge Calvo-Zaragoza. 2018. Optical Music Recognition in Mensural Notation with Region-based Convolutional Neural Networks. In *19th International Society for Music Information Retrieval Conference.* (in press).

[14] Alexander Pacha, Kwon-Young Choi, Bertrand Coüasnon, Yann Ricquebourg, Richard Zanibbi, and Horst Eidenberger. 2018. Handwritten Music Object Detection: Open Issues and Baseline Results. In *13th IAPR Workshop on Document Analysis Systems.* 163–168.

[15] Laurent Pugin. 2009. Editing Renaissance Music: The Aruspix Project. *Internationales Jahrbuch für Editionswissenschaften* (2009), 94–103.

[16] Laurent Pugin, Rodolfo Zitellini, and Perry Roland. 2014. Verovio - A library for Engraving MEI Music Notation into SVG.. In *International Society for Music Information Retrieval.*

[17] David Rizo, Pascual Beatriz, José Manuel Iñesta, Ezquerro, and Luis Antonio González Marín. 2017. Towards the Digital Encoding of Hispanic White Mensural Notation. *Anuario Musical* 72 (2017), 293–304.

[18] Eleanor Selfridge-Field. 1997. *Beyond MIDI: The handbook of musical codes.* MIT Press, Cambridge, Massachusetts, USA.

[19] David M. Weigl, Kevin R. Page, Peter Organisciak, and J. Stephen Downie. 2017. Information-seeking in Large-scale Digital Libraries: Strategies for Scholarly Workset Creation. In *Proceedings of the 17th ACM/IEEE Joint Conference on Digital Libraries (JCDL '17).* IEEE Press, Piscataway, NJ, USA, 253–256. http://dl.acm.org/citation.cfm?id=3200334.3200365